Proj

2026-04-25

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.1.6
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.2     ✓ tibble    3.3.1
## ✓ lubridate 1.9.4     ✓ tidyr     1.3.2
## ✓ purrr     1.2.1
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to be
come errors
```

```
library(COVID19)
```

```
## Warning: package 'COVID19' was built under R version 4.5.3
```

```
df <- covid19(level = 2)
```

```
## We have invested a lot of time and effort in creating COVID-19 Data
## Hub, please cite the following when using it:
##
##   Guidotti, E., Ardia, D., (2020), "COVID-19 Data Hub", Journal of Open
##   Source Software 5(51):2376, doi: 10.21105/joss.02376
##
## The implementation details and the latest version of the data are
## described in:
##
##   Guidotti, E., (2022), "A worldwide epidemiological database for
##   COVID-19 at fine-grained spatial resolution", Sci Data 9(1):112, doi:
##   10.1038/s41597-022-01245-1
## To print citations in BibTeX format use:
##  > print(citation('COVID19'), bibtex=TRUE)
##
## To hide this message use 'verbose = FALSE'.
```

```
colnames(df)
```

```
##  [1] "id"                                "date"
##  [3] "confirmed"                         "deaths"
##  [5] "recovered"                         "tests"
##  [7] "vaccines"                          "people_vaccinated"
##  [9] "people_fully_vaccinated"           "hosp"
## [11] "icu"                               "vent"
## [13] "school_closing"                    "workplace_closing"
## [15] "cancel_events"                     "gatherings_restrictions"
## [17] "transport_closing"                 "stay_home_restrictions"
## [19] "internal_movement_restrictions"    "international_movement_restrictions"
## [21] "information_campaigns"             "testing_policy"
## [23] "contact_tracing"                   "facial_coverings"
## [25] "vaccination_policy"                "elderly_people_protection"
## [27] "government_response_index"         "stringency_index"
## [29] "containment_health_index"          "economic_support_index"
## [31] "administrative_area_level"         "administrative_area_level_1"
## [33] "administrative_area_level_2"       "administrative_area_level_3"
## [35] "latitude"                          "longitude"
## [37] "population"                        "iso_alpha_3"
## [39] "iso_alpha_2"                       "iso_numeric"
## [41] "iso_currency"                      "key_local"
## [43] "key_google_mobility"               "key_apple_mobility"
## [45] "key_jhu_csse"                      "key_nuts"
## [47] "key_gadm"
```

```
df_clean <- df %>%
  filter(iso_alpha_3 == "USA") %>%
  filter(!is.na(administrative_area_level_2)) %>%
  select(
    date,
    state = administrative_area_level_2,
    deaths,
    confirmed,
    population
  )
```

```
df_cleaner <- df_clean %>%
  filter(!state %in% c("Puerto Rico","Northern Mariana Islands", "Virgin Islands", "American
Samoa"))

head(df_cleaner)
```

```
##         date     state deaths confirmed population
## 1 2020-01-01 Minnesota     NA        NA    5639632
## 2 2020-01-02 Minnesota     NA        NA    5639632
## 3 2020-01-03 Minnesota     NA        NA    5639632
## 4 2020-01-04 Minnesota     NA        NA    5639632
## 5 2020-01-05 Minnesota     NA        NA    5639632
## 6 2020-01-06 Minnesota     NA        NA    5639632
```

```
library(tidyverse)

elections <- tribble(
  ~state, ~margin_2016, ~margin_2020,
  "alabama", -27.7, -25.5,
  "alaska", -14.7, -10.1,
  "arizona", -3.5, 0.3,
  "arkansas", -26.9, -27.6,
  "california", 30.1, 29.2,
  "colorado", 4.9, 13.5,
  "connecticut", 13.6, 20.1,
  "delaware", 11.4, 19.0,
  "florida", -1.2, -3.4,
  "georgia", -5.1, 0.2,
  "hawaii", 32.2, 29.5,
  "idaho", -31.8, -30.8,
  "illinois", 17.1, 16.9,
  "indiana", -19.2, -16.1,
  "iowa", -9.4, -8.2,
  "kansas", -20.6, -14.7,
  "kentucky", -29.8, -25.9,
  "louisiana", -19.6, -18.6,
  "maine", 2.9, 9.1,
  "maryland", 26.4, 33.2,
  "massachusetts", 27.2, 33.5,
  "michigan", -0.2, 2.8,
  "minnesota", 1.5, 7.1,
  "mississippi", -17.8, -16.5,
  "missouri", -18.5, -15.4,
  "montana", -20.2, -16.4,
  "nebraska", -25.0, -19.1,
  "nevada", 2.4, 2.4,
  "new hampshire", 0.4, 7.4,
  "new jersey", 14.1, 15.9,
  "new mexico", 8.2, 10.8,
  "new york", 21, 23.1,
  "north carolina", -3.7, -1.3,
  "north dakota", -35.7, -33.4,
  "ohio", -8.1, -8.0,
  "oklahoma", -36.4, -33.1,
  "oregon", 11.0, 16.1,
  "pennsylvania", -0.7, 1.2,
  "rhode island", 15.5, 20.8,
  "south carolina", -14.3, -11.7,
  "south dakota", -29.8, -26.2,
  "tennessee", -26.0, -23.2,
  "texas", -9.0, -5.6,
  "utah", -18.1, -20.5,
  "vermont", 26.4, 35.4,
  "virginia", 5.3, 10.1,
  "washington", 16.2, 19.2,
```

```
  "west virginia", -42.1, -38.9,
  "wisconsin", -0.8, 0.6,
  "wyoming", -46.3, -43.4
)

elections_change <- elections %>%
  mutate(
    shift_16_20 = margin_2020 - margin_2016,
    shift_cat = case_when(
      shift_16_20 > 1 ~ "Shifted Democratic >1%",
      shift_16_20 < -1 ~ "Shifted Republican >1%",
      TRUE ~ "Small Change"
    )
  )
change_map_df <- map_data("state") %>%
  left_join(elections_change, by = c("region" = "state"))
```

```
library(lubridate)
library(maps)
```

```
##
## Attaching package: 'maps'
```

```
## The following object is masked from 'package:purrr':
##
##     map
```

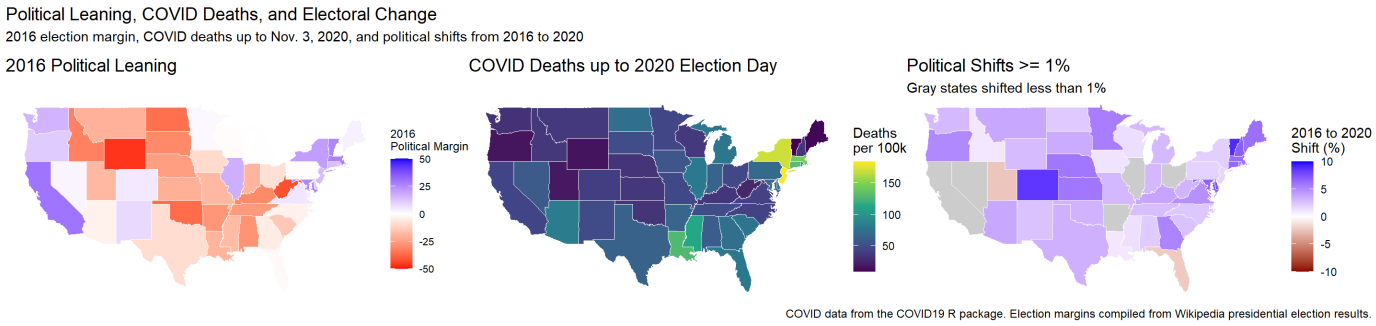
```
library(patchwork)

covid_2020_election <- df_cleaner %>%
  filter(date <= as.Date("2020-11-03")) %>%
  group_by(state) %>%
  slice_max(date, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  mutate(
    state = tolower(state),
    deaths_per_100k = deaths / population * 100000
  ) %>%
  select(state, date, deaths, population, deaths_per_100k)
```

```
states_map <- map_data("state")

map_df <- states_map %>%
  left_join(elections, by = c("region" = "state")) %>%
  left_join(covid_2020_election, by = c("region" = "state"))
```

```
map_2016 <- ggplot(map_df, aes(long, lat, group = group)) +
  geom_polygon(aes(fill = margin_2016), color = "white", linewidth = 0.2) +
  coord_fixed(1.3) +
  scale_fill_gradient2(
    low = "red",
    mid = "white",
    high = "blue",
    midpoint = 0,
    limits = c(-50, 50),   # makes legend consistent and readable
    name = "2016\nPolitical Margin"
  ) +
  labs(title = "2016 Political Leaning") +
  theme_void() +
  theme(
    legend.position = "right",
    legend.title = element_text(size = 9),
    legend.text = element_text(size = 8)
  )
map_deaths <- ggplot(map_df, aes(long, lat, group = group)) +
  geom_polygon(aes(fill = deaths_per_100k), color = "white", linewidth = 0.2) +
  coord_fixed(1.3) +
  scale_fill_viridis_c() +
  labs(
    title = "COVID Deaths up to 2020 Election Day",
    fill = "Deaths\nper 100k"
  ) +
  theme_void()


map_2020 <- ggplot(map_df, aes(long, lat, group = group)) +
  geom_polygon(aes(fill = margin_2020), color = "white", linewidth = 0.2) +
  coord_fixed(1.3) +
  scale_fill_gradient2(
    low = "red",
    mid = "white",
    high = "blue",
    midpoint = 0
  ) +
  labs(
    title = "2020 Election Margin",
    fill = "Democratic\nMargin"
  ) +
  theme_void()

shift_map_df <- change_map_df %>%
  mutate(
    shift_display = if_else(abs(shift_16_20) >= 1, shift_16_20, NA_real_)
  )

map_shift <- ggplot(shift_map_df, aes(long, lat, group = group)) +
  geom_polygon(aes(fill = shift_display), color = "white", linewidth = 0.2) +
```

```
  coord_fixed(1.3) +
scale_fill_gradient2(
  low = "darkred",
  mid = "white",
  high = "blue",
  midpoint = 0,
  limits = c(-10, 10),
  na.value = "gray80",
  name = "2016 to 2020\nShift (%)"
) +
  labs(
    title = "Political Shifts >= 1%",
    subtitle = "Gray states shifted less than 1%"
  ) +
  theme_void()
```

```
final_plot <- map_2016 + map_deaths + map_shift +
  plot_layout(ncol = 3) +
  plot_annotation(
    title = "Political Leaning, COVID Deaths, and Electoral Change",
    subtitle = "2016 election margin, COVID deaths up to Nov. 3, 2020, and political shifts f
rom 2016 to 2020",
    caption = "COVID data from the COVID19 R package. Election margins compiled from Wikipedi
a presidential election results."
  ) &
  theme(
    plot.title = element_text(size = 14),
    plot.subtitle = element_text(size = 11),
    plot.caption = element_text(size = 9)
  )

final_plot
```



```
df_cleaner %>%
  filter(state == "Maryland") %>%
  arrange(date) %>%
  select(date, deaths) %>%
  head(10)
```

```
##          date deaths
## 1  2020-03-01     NA
## 2  2020-03-02     NA
## 3  2020-03-03     NA
## 4  2020-03-04     NA
## 5  2020-03-05      0
## 6  2020-03-06      0
## 7  2020-03-07      0
## 8  2020-03-08      0
## 9  2020-03-09      0
## 10 2020-03-10      0
```

```
df_cleaner %>%
  filter(state == "Maryland") %>%
  arrange(date) %>%
  select(date, deaths) %>%
  tail(10)
```

```
##            date deaths
## 1540 2024-05-18     NA
## 1541 2024-05-19     NA
## 1542 2024-05-20     NA
## 1543 2024-05-21     NA
## 1544 2024-05-22     NA
## 1545 2024-05-23     NA
## 1546 2024-05-24     NA
## 1547 2024-05-25     NA
## 1548 2024-05-26     NA
## 1549 2024-05-27     NA
```

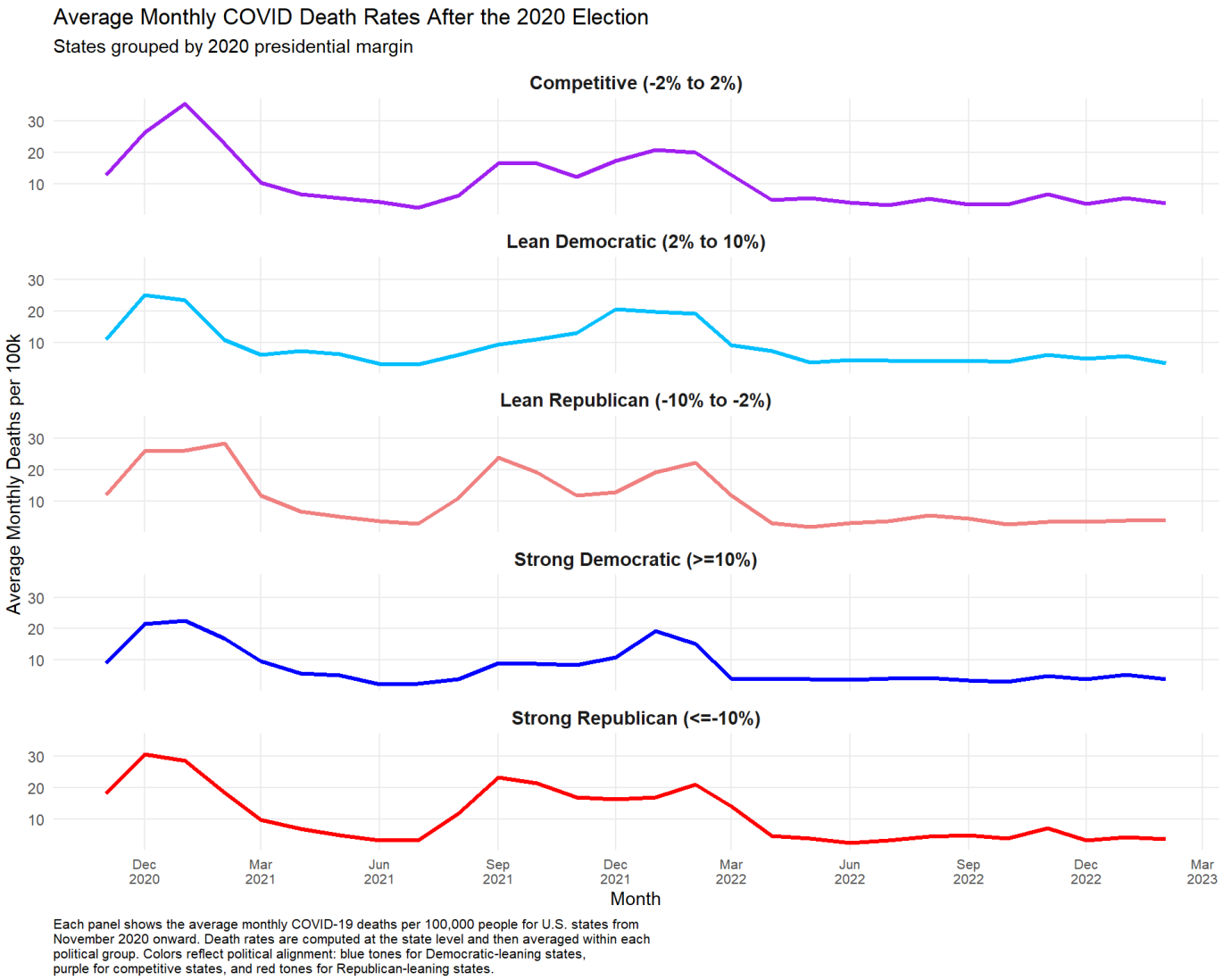
```
election_day <- as.Date("2020-11-03")
```

```
# Get each state's cumulative deaths on Election Day
baseline_deaths <- df_cleaner %>%
  filter(date <= election_day) %>%
  group_by(state) %>%
  slice_max(date, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  transmute(
    state = tolower(state),
    baseline_deaths = deaths,
    population = population
  )
```

```
# Monthly cumulative deaths after Election Day
monthly_deaths <- df_cleaner %>%
  filter(date > election_day) %>%
  mutate(
    state = tolower(state),
    month = floor_date(date, "month")
  ) %>%
  group_by(state, month) %>%
  slice_max(date, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  select(state, month, deaths)
```

```
# Convert cumulative deaths into deaths added each month
monthly_analysis <- monthly_deaths %>%
  left_join(baseline_deaths, by = "state") %>%
  left_join(elections, by = "state") %>%
  arrange(state, month) %>%
  group_by(state) %>%
  mutate(
    monthly_deaths = deaths - lag(deaths, default = first(baseline_deaths)),
    monthly_deaths_per_100k = monthly_deaths / population * 100000,
    margin_bracket = case_when(
      margin_2020 <= -10 ~ "Strong Republican",
      margin_2020 > -10 & margin_2020 <= -2 ~ "Lean Republican",
      margin_2020 > -2 & margin_2020 < 2 ~ "Competitive",
      margin_2020 >= 2 & margin_2020 < 10 ~ "Lean Democratic",
      margin_2020 >= 10 ~ "Strong Democratic"
    )
  ) %>%
  ungroup()
```

```
monthly_summary <- monthly_analysis %>%
  filter(!is.na(margin_bracket)) %>%
  group_by(month, margin_bracket) %>%
  summarise(
    avg_deaths_per_100k = mean(monthly_deaths_per_100k, na.rm = TRUE),
    .groups = "drop"
  )
library(stringr)

party_colors <- c(
  "Strong Democratic (>=10%)" = "blue",
  "Lean Democratic (2% to 10%)" = "deepskyblue",
  "Competitive (-2% to 2%)" = "purple",
  "Lean Republican (-10% to -2%)" = "lightcoral",
  "Strong Republican (<=-10%)" = "red"
)


monthly_summary_plot <- monthly_summary %>%
  mutate(
    margin_label = recode(margin_bracket,
      "Strong Democratic" = "Strong Democratic (>=10%)",
      "Lean Democratic"   = "Lean Democratic (2% to 10%)",
      "Competitive"       = "Competitive (-2% to 2%)",
      "Lean Republican"   = "Lean Republican (-10% to -2%)",
      "Strong Republican" = "Strong Republican (<=-10%)"
    )
  )

caption_text <- str_wrap(
  "Each panel shows the average monthly COVID-19 deaths per 100,000 people for U.S. states fr
om November 2020 onward. Death rates are computed at the state level and then averaged within
each political group. Colors reflect political alignment: blue tones for Democratic-leaning s
tates, purple for competitive states, and red tones for Republican-leaning states.",
  width = 100
)


ggplot(monthly_summary_plot,
       aes(x = month, y = avg_deaths_per_100k, color = margin_label)) +
  geom_line(linewidth = 1.2) +
  facet_wrap(~margin_label, ncol = 1) +
  scale_color_manual(values = party_colors) +
  scale_x_date(
  limits = c(as.Date("2020-11-01"), as.Date("2023-02-01")),
  date_breaks = "3 months",
  date_labels = "%b\n%Y"
) +
  labs(
    title = "Average Monthly COVID Death Rates After the 2020 Election",
    subtitle = "States grouped by 2020 presidential margin",
```

```
    x = "Month",
    y = "Average Monthly Deaths per 100k",
    caption = caption_text
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(size = 8),
    panel.grid.minor = element_blank(),
    strip.text = element_text(size = 11, face = "bold"),
    legend.position = "none",
    plot.caption = element_text(size = 8, hjust = 0)
  )
```

```
## Warning: Removed 76 rows containing missing values or values outside the scale range
## (`geom_line()`).
```